

MUSKRAT DAM, ON, CanadaWMO: **716780**Lat: **53.440N**Lon: **91.760W**Elev: **911**StdP: **14.22**Time Zone: **-6.00 (NAC)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.0	-30.1	-42.3	0.5	-34.6	-37.3	0.7	-29.9	20.4	4.0	18.4	3.9	3.6	220	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	81.3	64.6	78.5	63.1	75.8	62.0	68.2	76.9	66.0	73.9	64.1	71.8	9.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	96.5	71.5	63.2	89.6	69.6	61.0	82.8	67.9	33.1	76.9	31.3	74.3	29.9	71.9	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.6	17.9	15.8	DB	-42.3	87.2	4.3	3.3	-45.4	89.6	-47.9	91.5	-50.3	93.3	-53.5	95.7	
			WB	-42.7	71.5	4.1	3.7	-45.6	74.2	-48.0	76.4	-50.3	78.5	-53.3	81.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.4	-4.4	-3.3	12.9	28.4	44.1	57.5	63.4	60.3	50.3	37.0	16.7	-0.5
	DBStd	26.86	14.01	12.82	14.21	13.21	10.11	7.63	6.17	6.39	8.84	8.65	13.50	14.89
	HDD50	8328	1686	1492	1152	650	241	20	1	3	104	416	998	1565
	HDD65	12740	2151	1912	1616	1098	649	245	103	173	446	869	1448	2030
	CDD50	1170	0	0	0	2	59	245	417	321	113	13	0	0
	CDD65	108	0	0	0	0	2	19	55	27	5	0	0	0
	CDH74	987	0	0	0	0	49	201	475	213	47	2	0	0
	CDH80	159	0	0	0	0	9	16	99	28	7	0	0	0
Wind	WSAvg	7.1	6.8	6.9	7.2	7.4	7.8	7.0	6.8	7.0	7.2	7.6	7.0	6.8
Precipitation	PrecAvg	22.0	0.8	0.7	0.9	1.0	1.9	2.6	3.7	3.3	2.7	2.0	1.5	0.9
	PrecMax	28.7	1.4	1.4	1.8	1.9	4.3	6.2	6.0	4.9	5.7	3.7	2.4	2.2
	PrecMin	17.9	0.3	0.2	0.2	0.3	0.7	1.2	1.5	1.7	0.7	0.5	0.6	0.2
	PrecStd	2.5	0.3	0.3	0.4	0.4	0.7	0.9	1.3	0.9	1.1	0.8	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	34.5	56.0	66.7	80.0	82.1	87.9	83.3	79.2	67.8	50.0	32.9
		MCWB	30.3	31.7	48.6	50.3	58.4	62.9	70.3	66.6	63.6	55.7	45.2	31.5
	2%	DB	28.4	28.7	45.8	59.3	72.8	79.2	82.4	79.2	73.7	59.9	42.8	28.6
		MCWB	27.6	26.5	37.7	45.8	55.6	61.8	66.6	65.1	61.7	51.9	38.8	27.8
	5%	DB	23.8	22.9	39.2	53.6	66.9	76.3	79.0	76.0	69.4	55.1	37.8	25.1
		MCWB	22.6	20.4	34.2	42.7	52.6	60.7	63.7	63.5	60.0	50.0	35.4	24.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.6	33.0	48.4	51.2	63.0	69.1	73.3	69.3	66.6	57.8	45.7	31.3
		MCDB	31.6	33.9	57.1	62.4	75.9	76.5	83.5	79.0	74.1	64.0	49.9	32.2
	2%	WB	27.7	26.4	38.7	46.9	57.3	65.0	69.2	67.0	63.6	53.4	39.9	27.8
		MCDB	28.5	28.8	44.4	58.4	69.4	72.5	78.2	75.3	70.8	58.4	41.9	28.5
	5%	WB	22.7	21.0	34.3	42.8	54.4	62.7	66.7	65.3	60.8	49.8	35.5	24.3
		MCDB	23.8	22.8	38.5	53.0	64.8	72.2	75.6	72.8	66.7	54.6	37.7	25.0
Mean Daily Temperature Range	5% DB	WB	17.0	15.7	31.0	39.2	51.3	60.5	64.6	63.3	58.1	46.2	32.4	20.4
		MCDB	17.8	17.7	34.3	46.6	61.3	69.3	72.1	71.0	64.7	50.8	33.9	21.0
		MCWBR	18.1	21.0	17.4	17.5	13.9	11.2	10.9	12.0	12.0	12.8	11.4	11.8
	5% WB	MDBR	18.4	23.2	25.5	24.2	20.8	20.5	19.7	18.9	16.9	11.6	13.1	15.3
		MCDBR	18.7	23.6	26.5	29.4	29.6	24.6	24.0	23.4	22.6	20.7	15.3	12.7
		MCWBR	18.0	21.1	19.9	18.2	15.5	11.4	11.4	11.6	12.3	13.2	12.1	12.0
Clear-Sky Solar Irradiance	taub	0.237	0.276	0.279	0.321	0.347	0.354	0.384	0.358	0.328	0.287	0.264	0.227	
	taud	2.280	2.199	2.292	2.369	2.432	2.448	2.397	2.468	2.518	2.589	2.404	2.284	
	Ebn at Noon	243	262	290	290	286	284	272	275	271	264	231	224	
	Edh at Noon	18	28	32	34	34	34	35	31	26	19	16	15	
All-Sky Solar Radiation	RadAvg	331	712	1191	1667	1674	1783	1706	1396	942	522	265	210	
	RadStd	34	45	89	126	148	178	152	117	123	57	30	31	